

Engineering Graphics

SITAMS CSE (AI & ML) • R23 • I Year • II Semester • Course Code: 23ESC113T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit 1: ENGINEERING CURVES AND SCALES

Syllabus topics

- Introduction: Introduction about lines, lettering and dimensioning – Geometrical constructions and constructing regular polygons by general methods
- Engineering Curves: Construction of ellipse, parabola and hyperbola by general method – Construction of cycloids – Construction involutes – Drawing of tangents and normal to the above curves
- Scales: Plain scales, and diagonal

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit 2: PROJECTION OF POINTS, LINES AND PLANE SURFACES

Syllabus topics

- Projection of Points: Principles of orthographic projection – Reference lines and Plane – Projections of points
- Projection of Lines: Projections of lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane – Determination of true lengths, true inclinations by rotating line and trapezoidal method
- Projection of Planes: Perpendicular to both reference planes, parallel to one reference plane and inclined to the other reference plane
- plane inclined to both the reference planes

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Unit 3: PROJECTION OF SOLIDS

Syllabus topics

- Projection of Solids: Types of solids – Projection of simple solids (prisms, pyramids, cylinder and cone) – Axis perpendicular to horizontal plane, axis perpendicular to vertical plane and axis parallel to both the reference planes, projection of solids with axis inclined to one reference plane and parallel to another plane

Revision checklist

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Unit 4: SECTION OF SOLIDS AND DEVELOPMENT OF SURFACES

Syllabus topics

- Section of Solids: Sectioning of right regular solids like prisms, pyramids, cylinder and cone – Solids in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other plane – Obtaining true shape of section
- Development of Surfaces: Development of Surfaces: Development of lateral surfaces of simple and sectioned solids like prisms, pyramids, cylinder and cone

Revision checklist

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Unit 5: ISOMETRIC PROJECTIONS AND ORTHOGRAPHIC PROJECTIONS

Syllabus topics

- Isometric Projection: Principles of isometric projection – Isometric scale – Isometric views of simple solids and truncated solids like prisms, pyramids, cylinder, sphere, and cone
- Orthographic Projections: Visualization principles – Plane of projections – Representation of three-dimensional objects – Sketching of multiple views from pictorial views – Conversion of isometric views to orthographic views – Conversion of orthographic views to isometric views
- COMPUTER GRAPHICS (Not for Examination) (3) Practicing of simple 2D and 3D drawings of objects using Auto CAD

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This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.